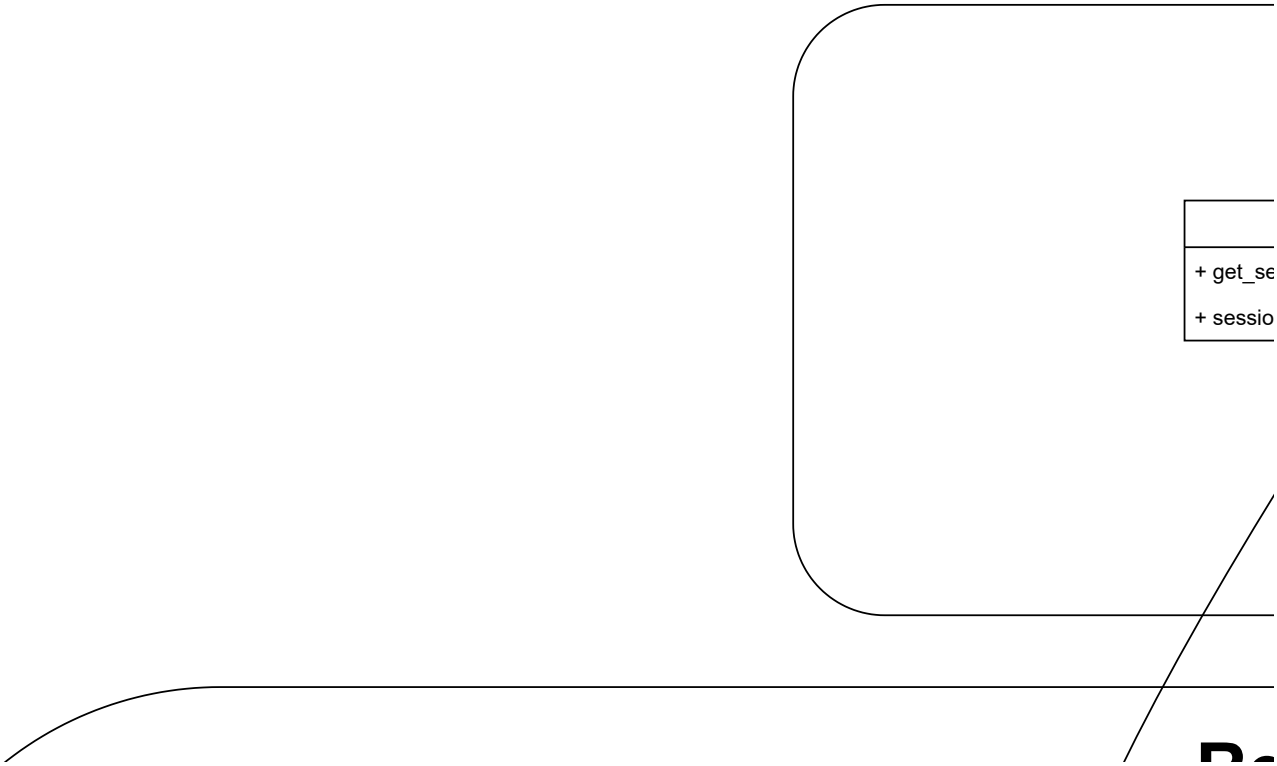
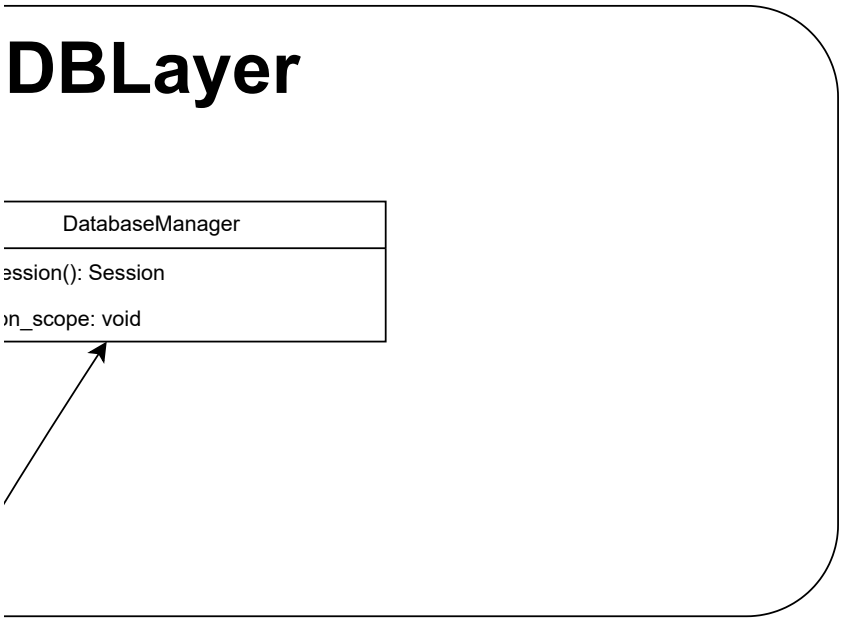
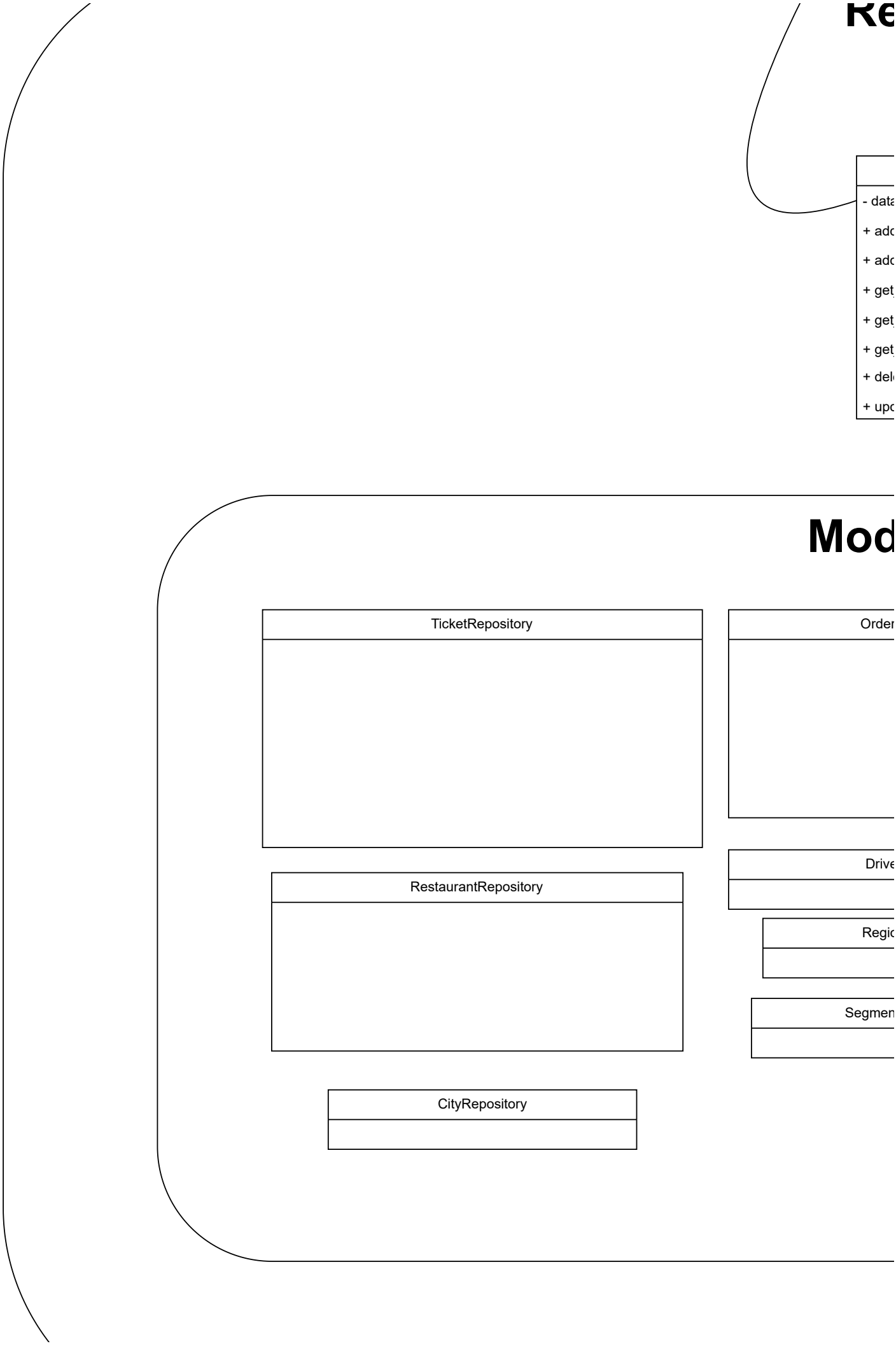
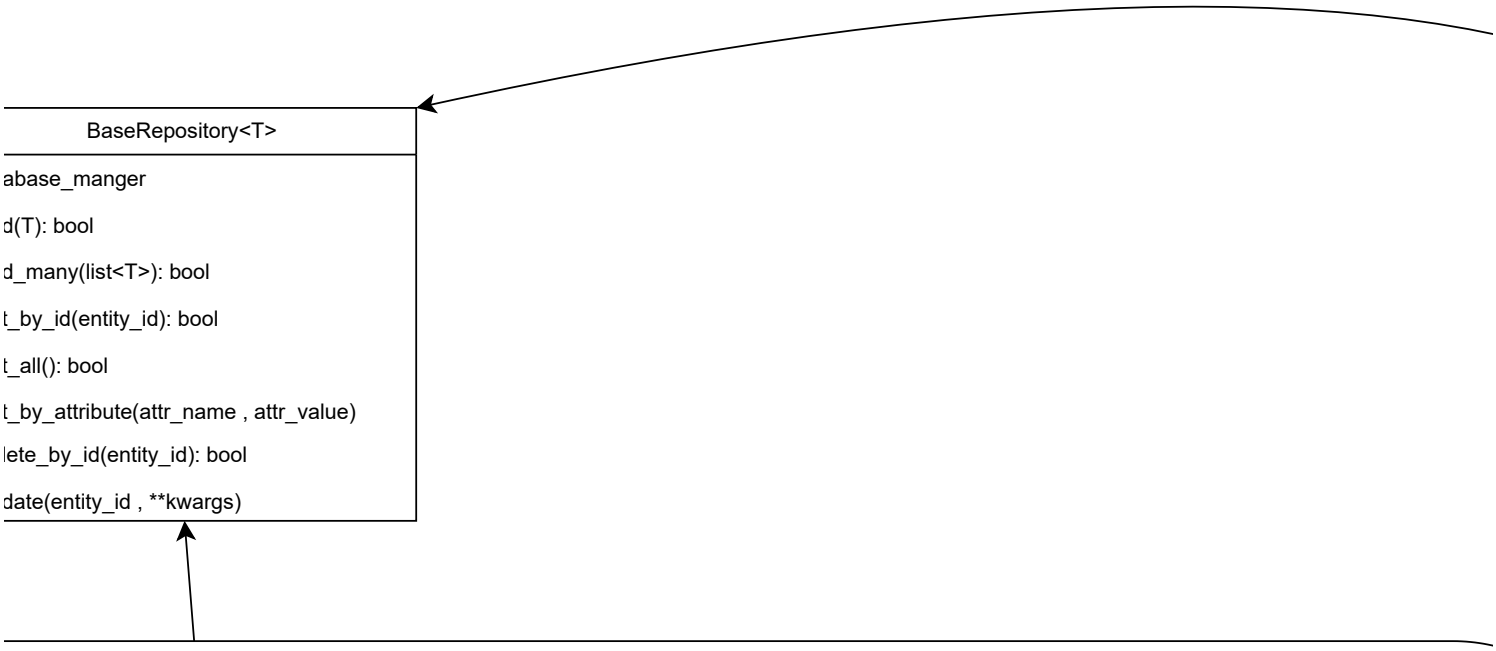

+ get_se
+ sessio

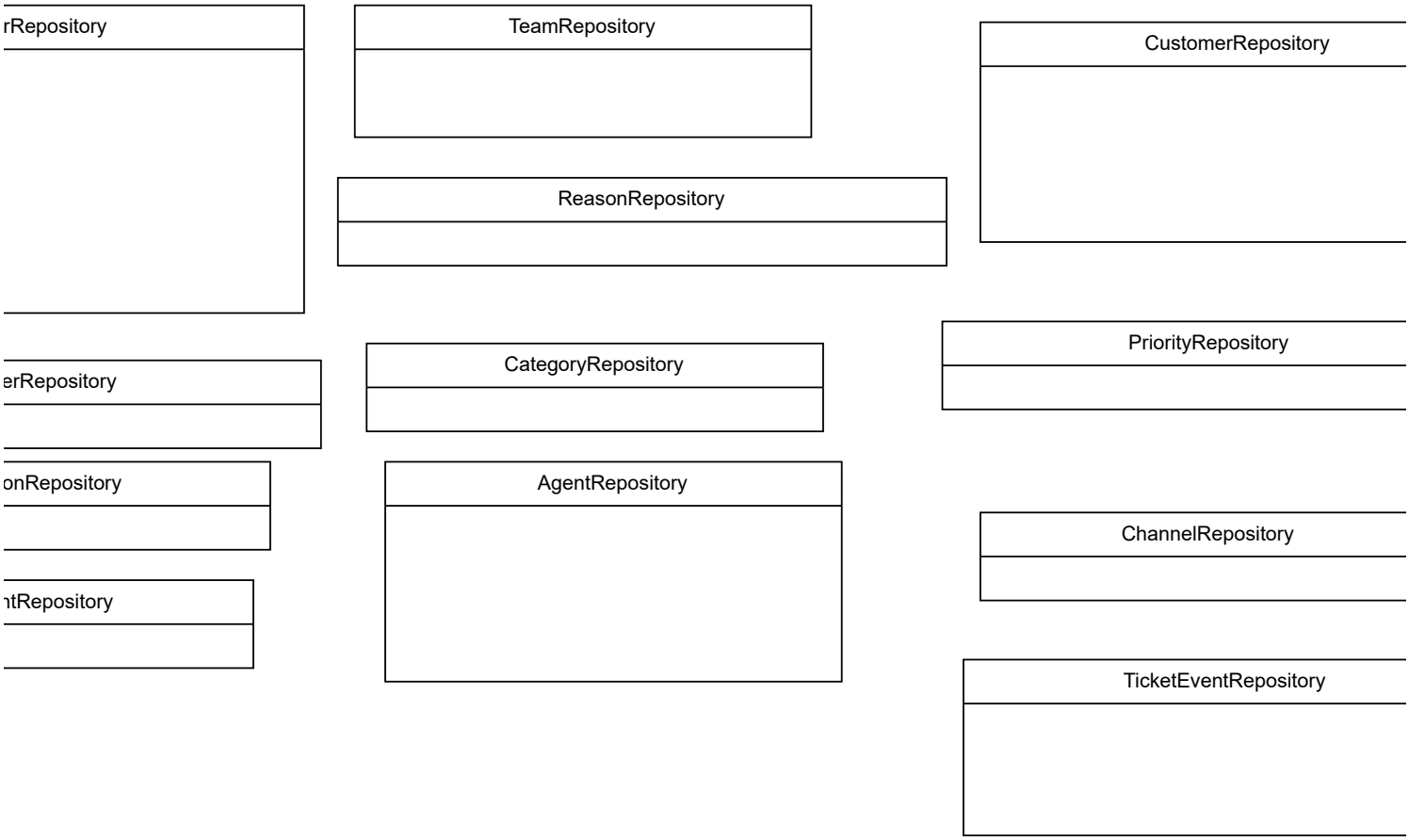

DBLayer
DatabaseManager
ession(): Session
on_scope: void

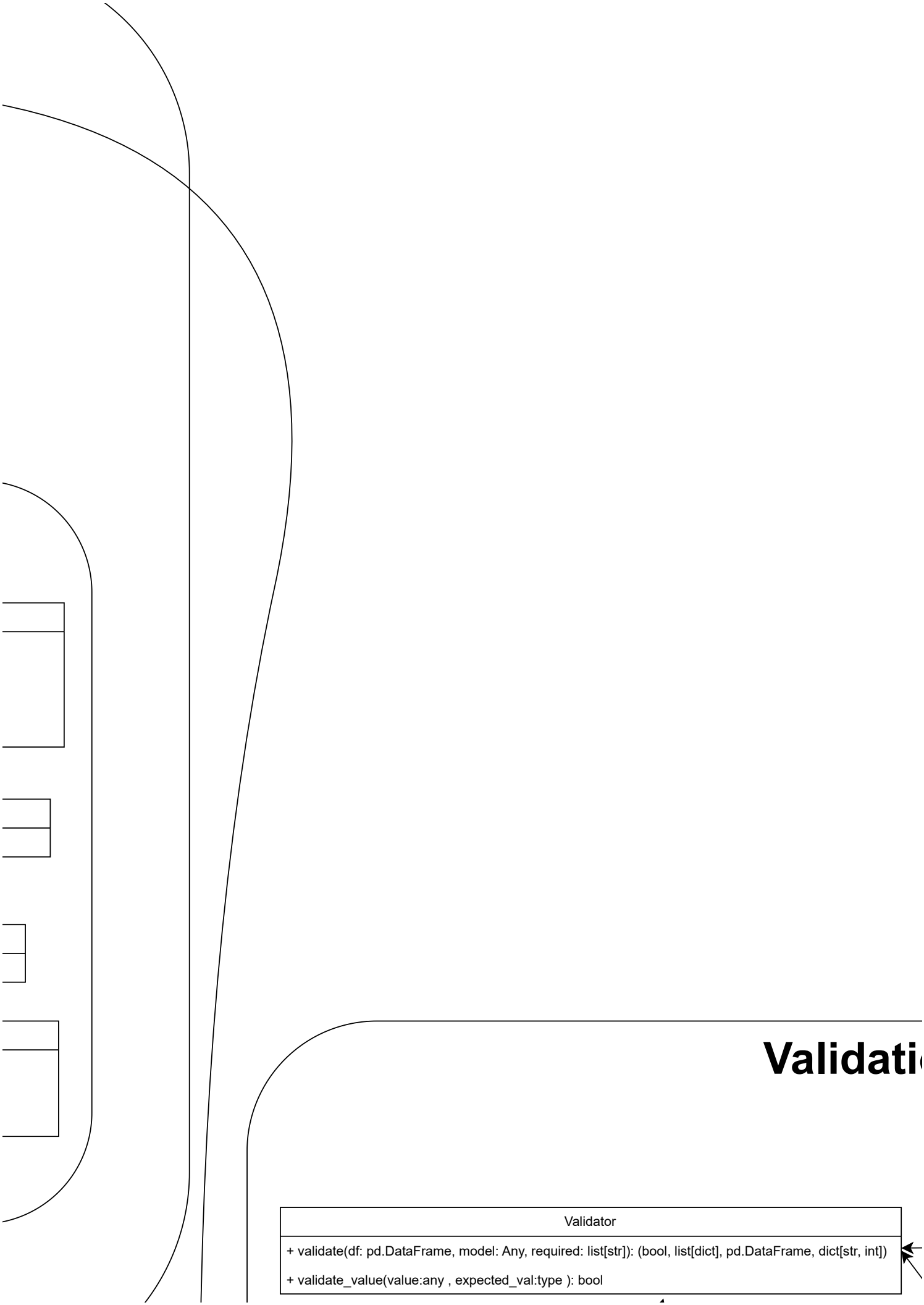

Re
- data
+ add
+ add
+ get
+ get
+ get
+ del
+ upd
Mod
TicketRepository
Order
Drive
RestaurantRepository
Regio
Segmen
CityRepository

epositoryLayer

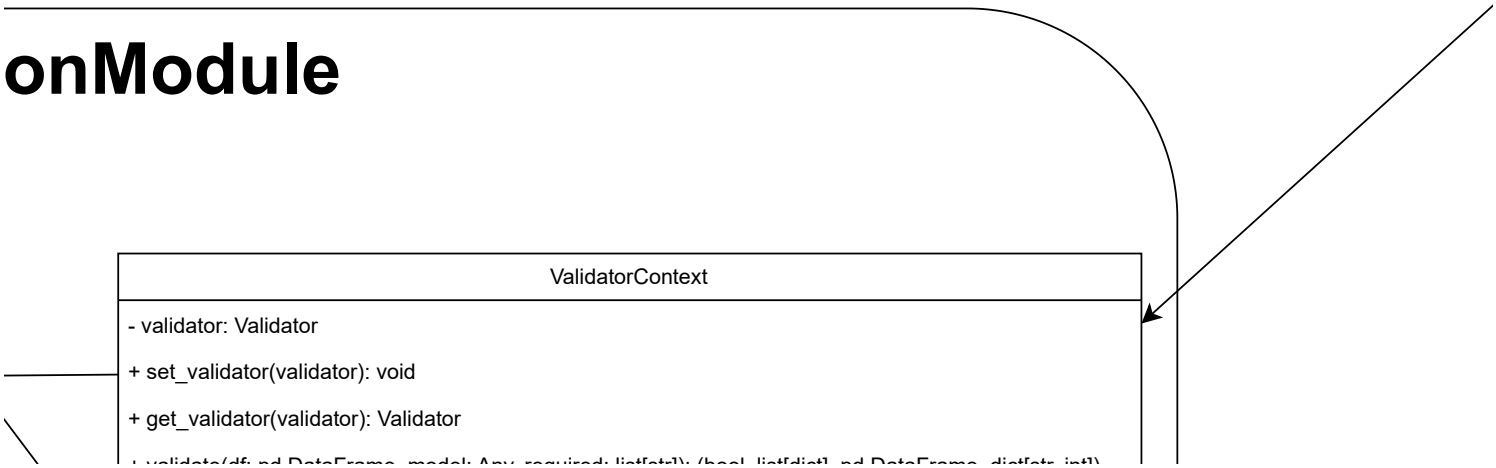


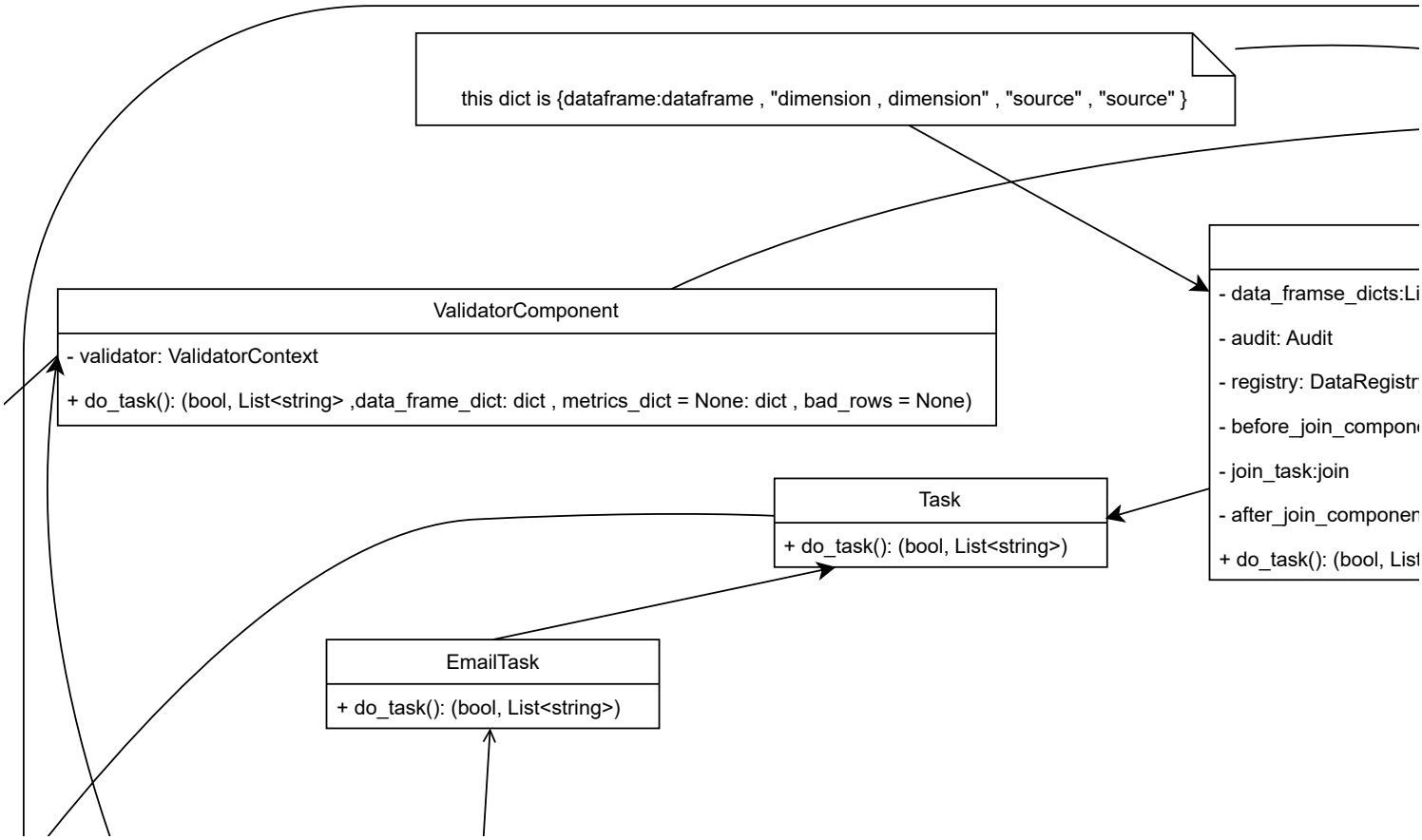
lelsRepository



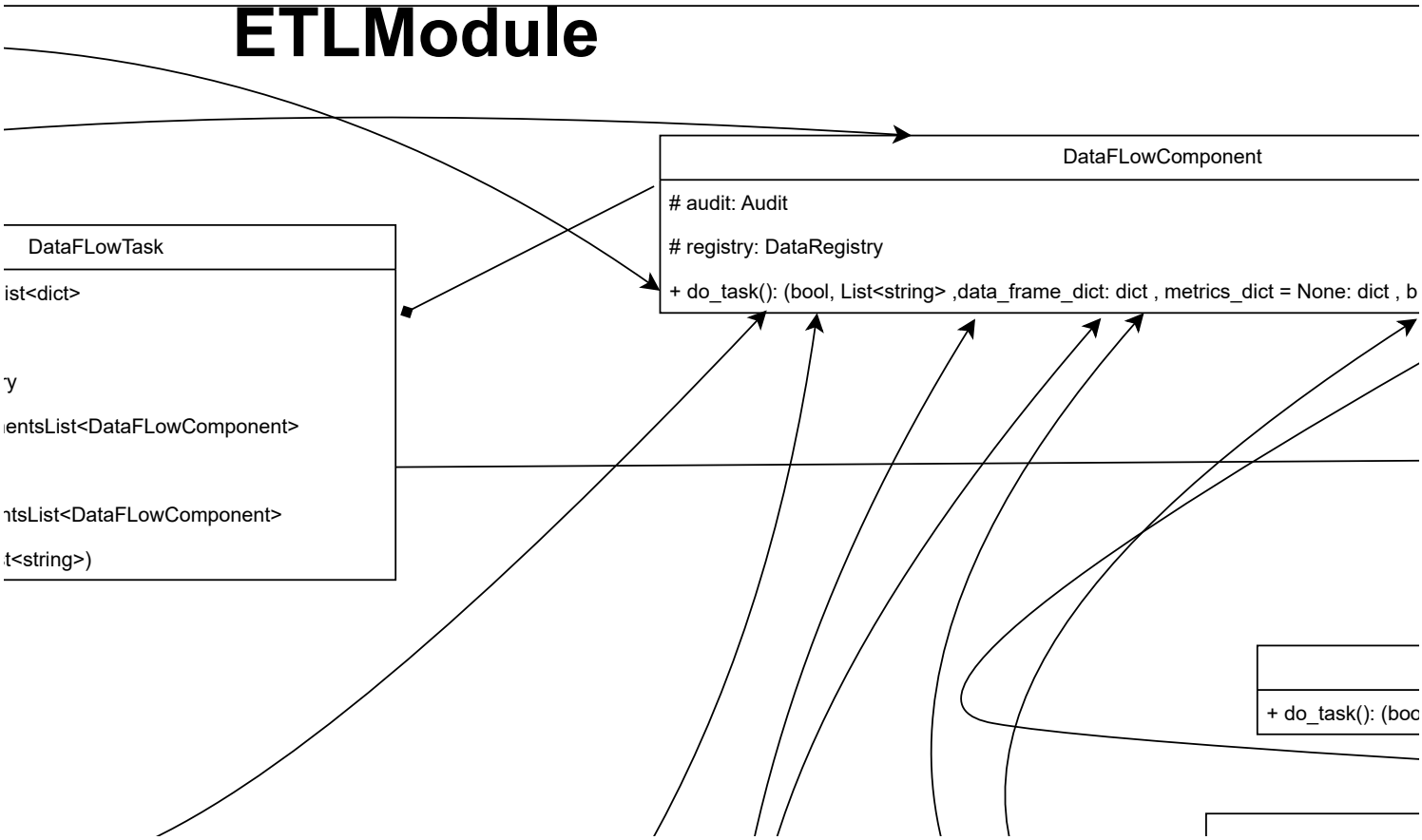

Validati
Validator
+ validate(df: pd.DataFrame, model: Any, required: list[str]): (bool, list[dict], pd.DataFrame, dict[str, int])
+ validate_value(value:any , expected_val:type): bool

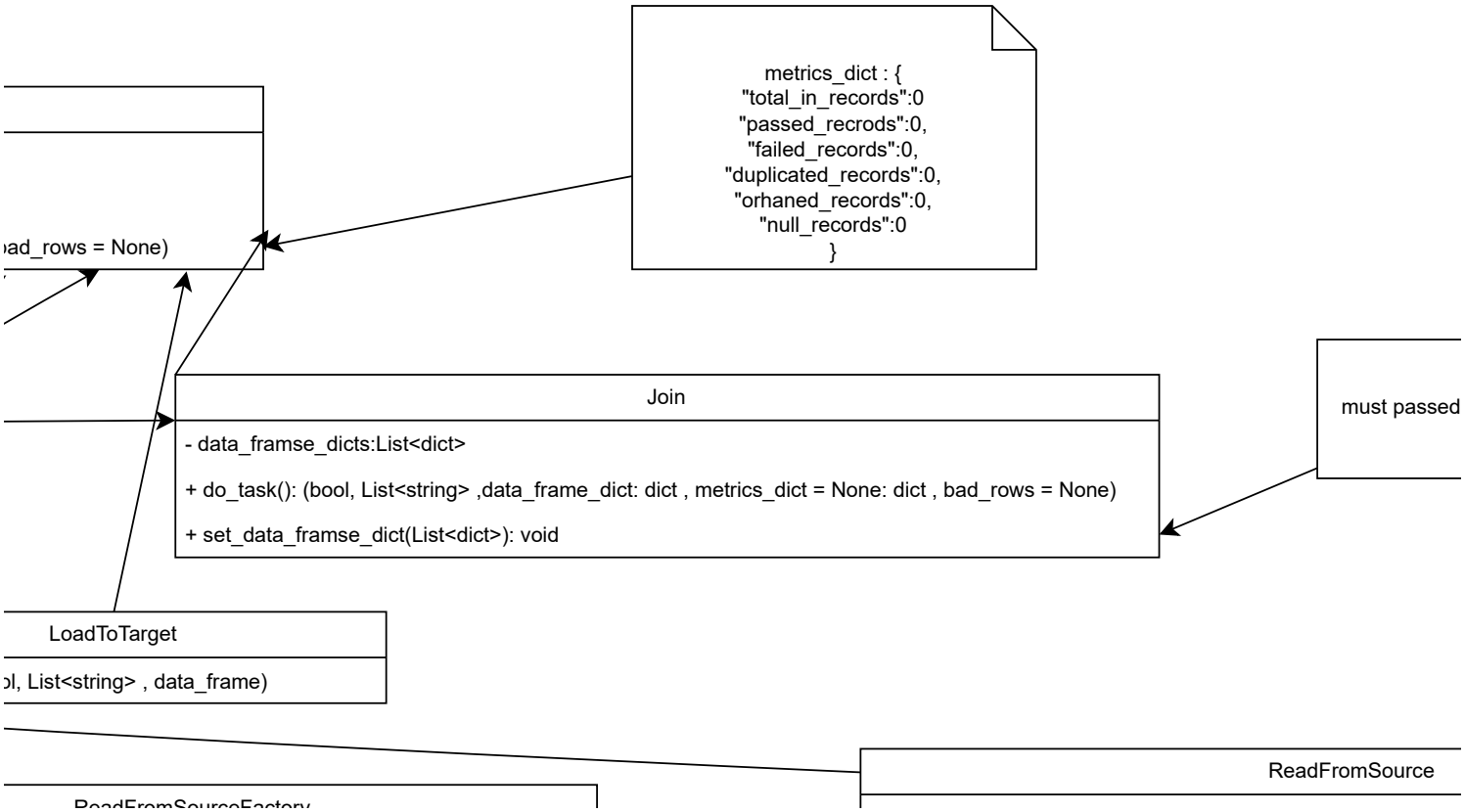
onModule

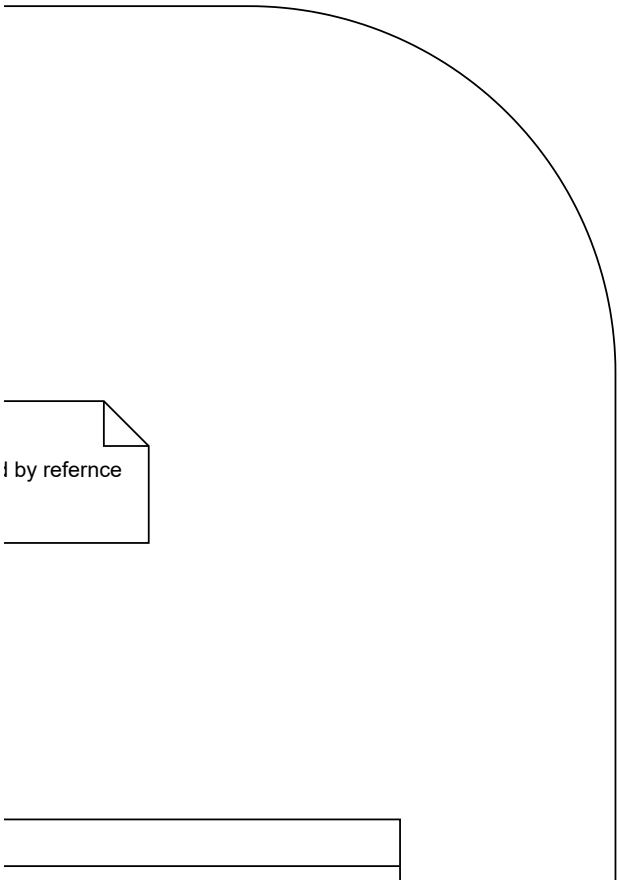

ValidatorContext
- validator: Validator
+ set_validator(validator): void
+ get_validator(validator): Validator


this dict is {dataframe:dataframe , "dimension , dimension" , "source" , "source" }
ValidatorComponent
- validator: ValidatorContext
+ do_task(): (bool, List<string> ,data_frame_dict: dict , metrics_dict = None: dict , bad_rows = None)
- data_framse_dicts:Li
- audit: Audit
- registry: DataRegistr
- before_join_compon
- join_task:join
- after_join_componen
+ do_task(): (bool, List
Task
+ do_task(): (bool, List<string>)
EmailTask
+ do_task(): (bool, List<string>)

ETLModule


DataFLowComponent
audit: Audit
registry: DataRegistry
+ do_task(): (bool, List<string> ,data_frame_dict: dict , metrics_dict = None: dict , b
DataFLowTask
ist<dict>
y
entsList<DataFLowComponent>
ntsList<DataFLowComponent>
t<string>)
+ do_task(): (boo


metrics_dict : {
"total_in_records":0
"passed_recrods":0,
"failed_records":0,
"duplicated_records":0,
"orhaned_records":0,
"null_records":0
}
ad_rows = None)
Join
- data_framse_dicts:List<dict>
+ do_task(): (bool, List<string> ,data_frame_dict: dict , metrics_dict = None: dict , bad_rows = None)
+ set_data_framse_dict(List<dict>): void
must passed
LoadToTarget
l, List<string> , data_frame)
ReadFromSource


l by refernce

Models

BaseModel

Entites

Ticket	
PK	**ticket_id : UUID NOT NULL**
FK	order_id : UUID NOT NULL
FK	customer_id : INT NOT NULL
FK	driver_id : INT NOT NULL
FK	restaurant_id : INT NOT NULL
FK	agent_id : INT NOT NULL
FK	reason_id : INT NOT NULL
FK	priority_id : INT NOT NULL
FK	channel_id : INT NOT NULL
	status : VARCHAR(8) NOT NULL
	refund_amount : DECIMAL NOT NULL
	created_at : TIMESTAMP NOT NULL
	first_response_at : TIMESTAMP NOT NULL
	resolved_at : TIMESTAMP NOT NULL
	sla_first_due_at : TIMESTAMP NOT NULL
	sla_resolve_due_at : TIMESTAMP NOT NULL

Agent	
PK	**agent_id : INT NOT NULL**
	agent_name : VARCHAR(10) NOT NULL

Team	
PK	**team_id : INT NOT NULL**
	team_name : VARCHAR(15) NC

ReasonCategory	
PK	**reason_category_id : INT NOT**
	category_name : VARCHAR(8) I

City	
PK	**city_id : INT NOT NULL**
	city_name : VARCHAR(10) NOT
	country : VARCHAR(5) NOT NU
	timezone : VARCHAR(12) NOT

Customer	
PK	**customer_id : INT NOT NULL**
	full_name : VARCHAR(20) NULI
	email : VARCHAR(22) NULL

ReasonCategoryRepository
+ get_by_name(category_name: str): ReasonCategory

OT NULL	

T NULL	
NOT NULL	

T NULL	
ULL	
NULL	

L	

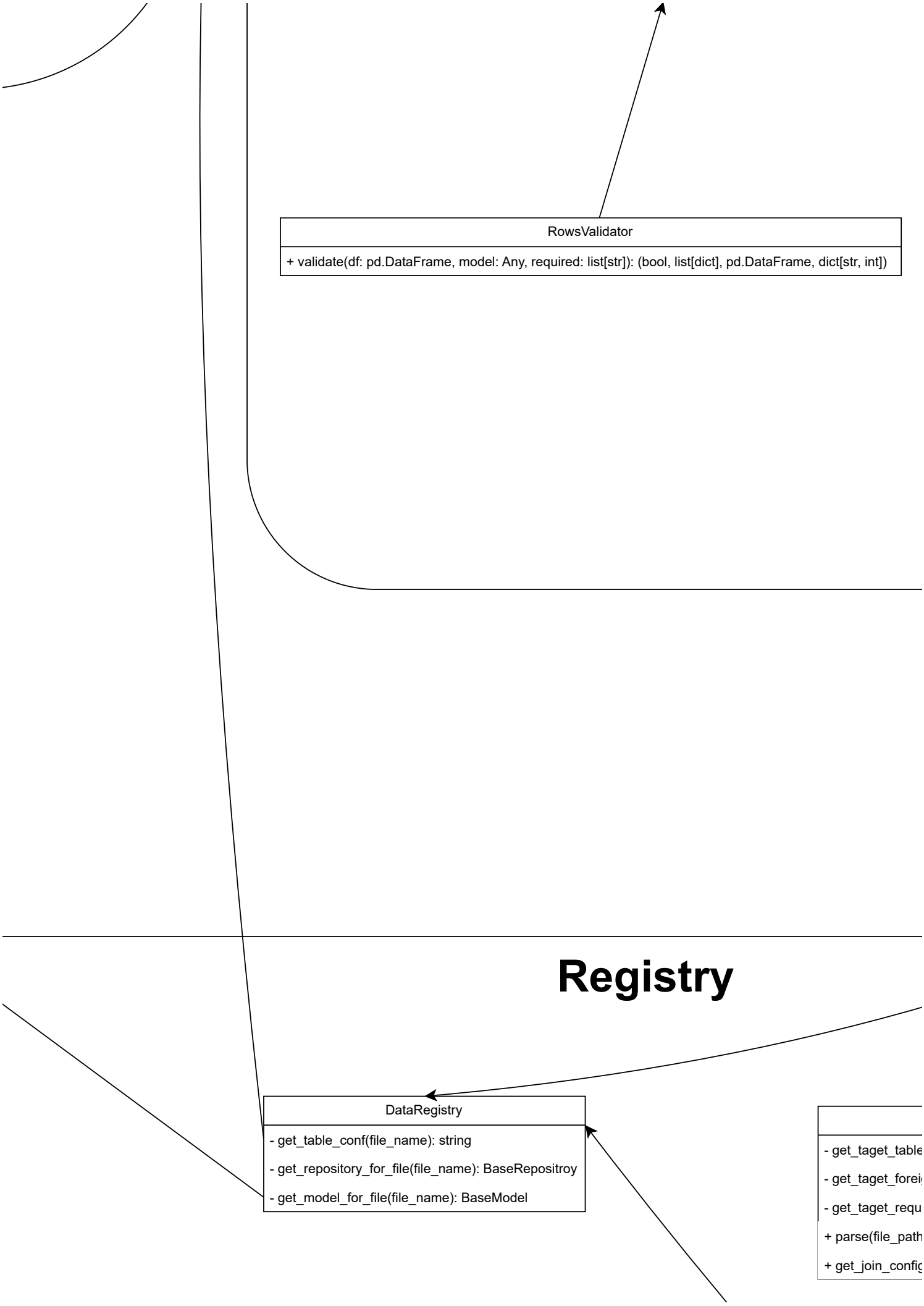
PK	**segment_id : INT NOT NULL**
	segment_name : VARCHAR(7) NOT NULL
	discount_pct : INT NOT NULL
	priority_support : BOOLEAN NOT NULL

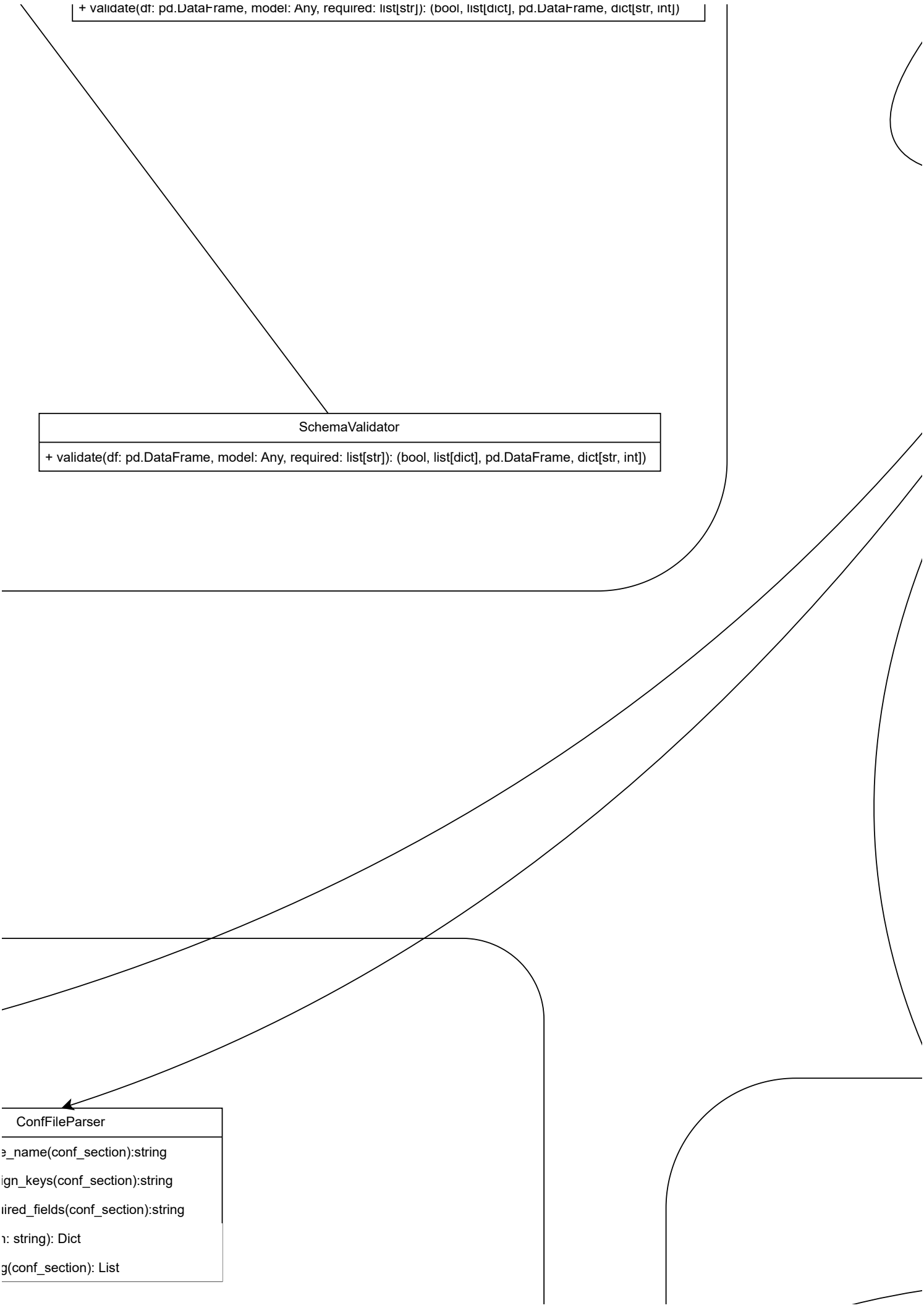
Segment

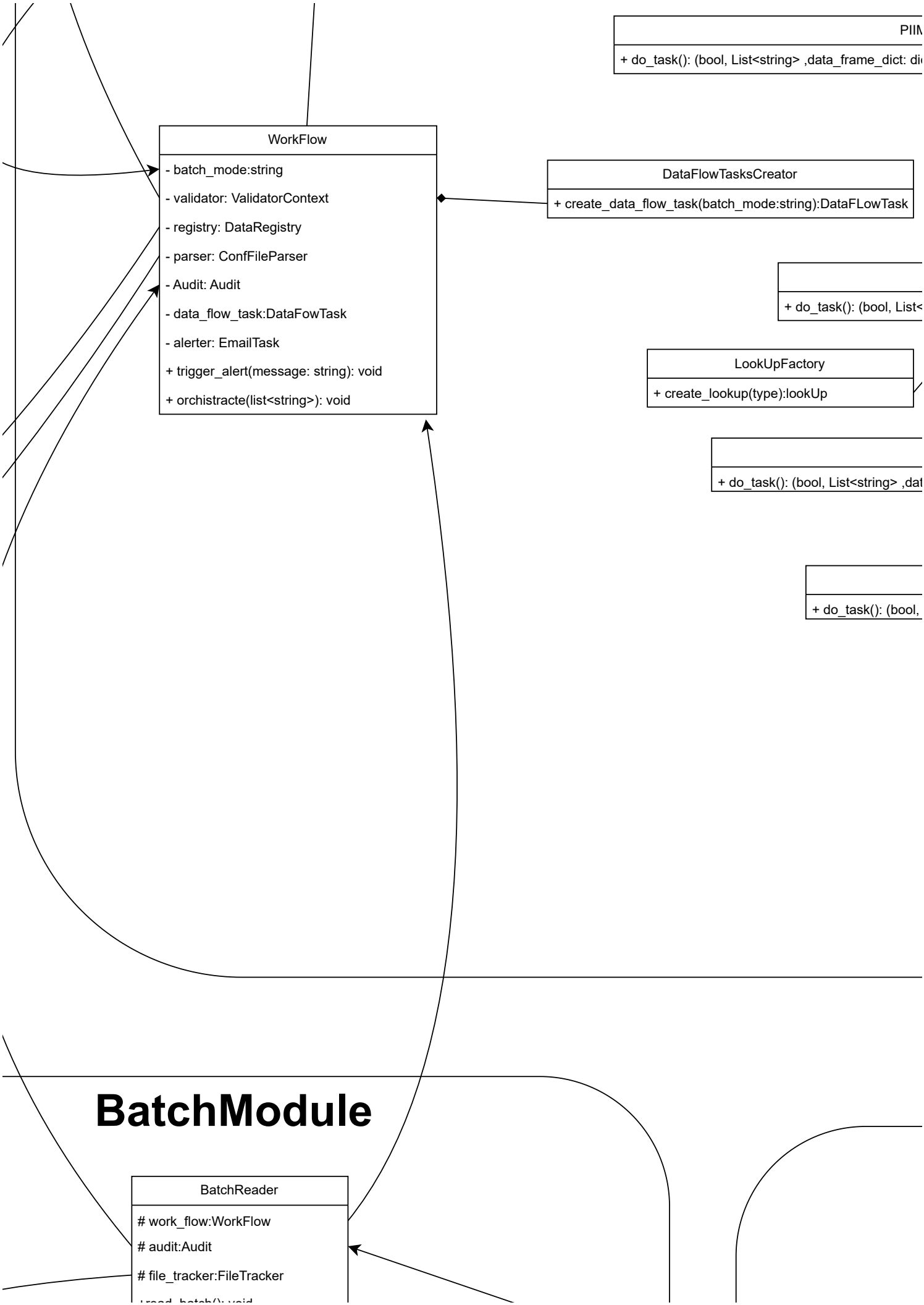
Priority	
PK	**priority_id : INT NOT NULL**
	priority_code : VARCHAR(2) NOT NULL
	priority_name : VARCHAR(8) NOT NULL
	sla_first_response_min : INT NOT NULL
	sla_resolution_min : INT NOT NULL

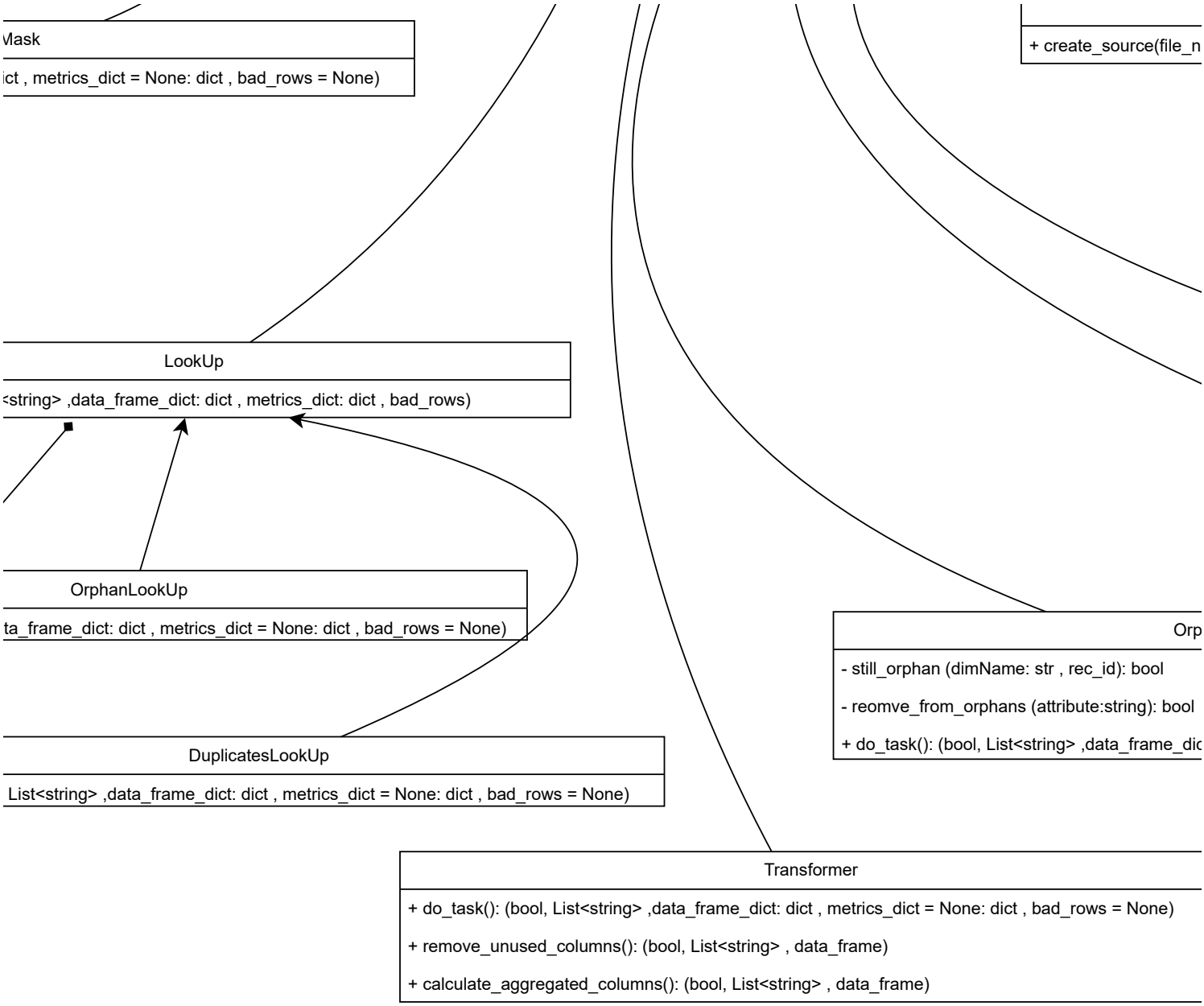
Channel	
PK	**channel_id : INT NOT NULL**
	channel_name : VARCHAR(5) NOT NULL

Order

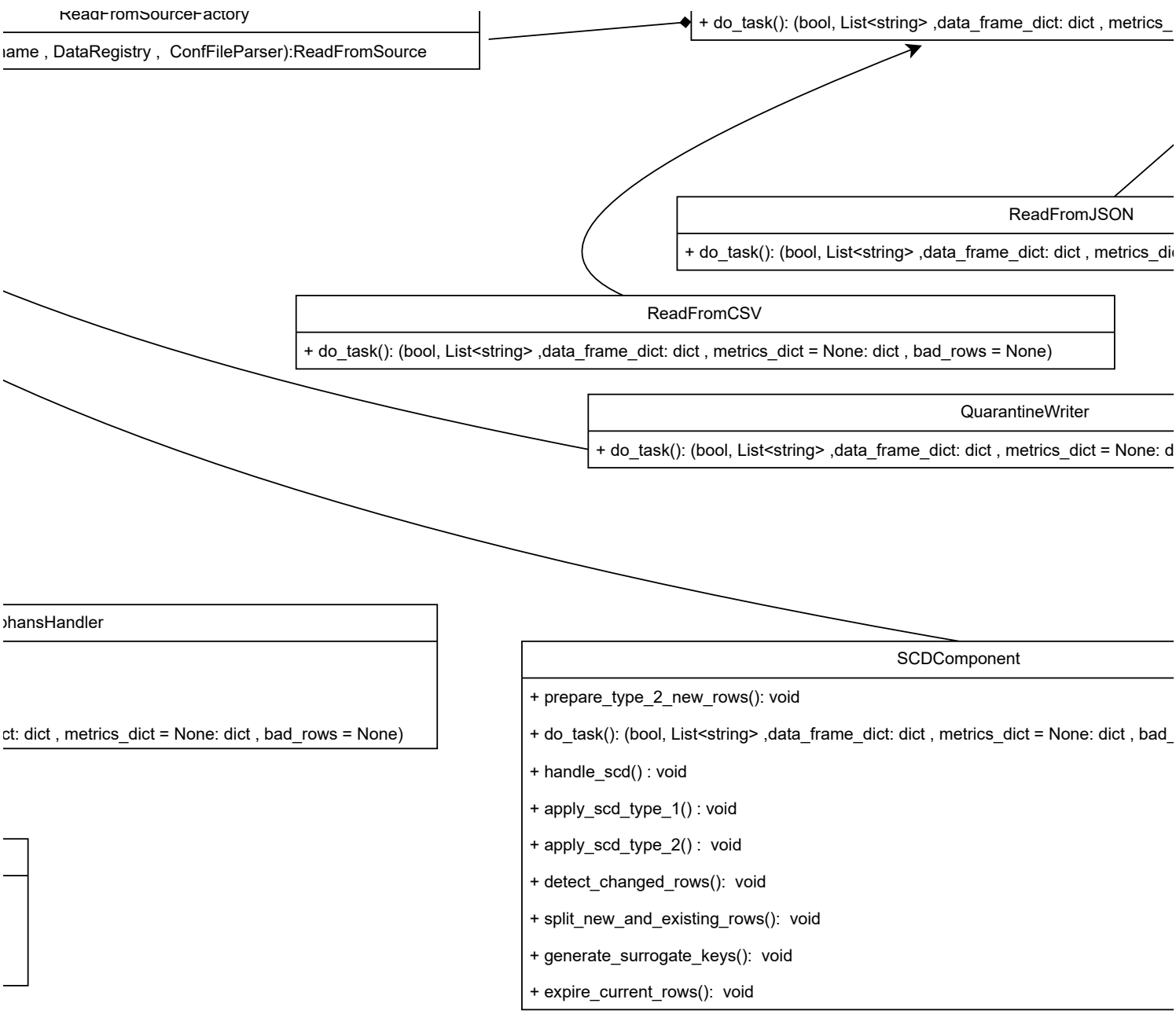

RowsValidator
+ validate(df: pd.DataFrame, model: Any, required: list[str]): (bool, list[dict], pd.DataFrame, dict[str, int])
Registry
DataRegistry
- get_table_conf(file_name): string
- get_repository_for_file(file_name): BaseRepositroy
- get_model_for_file(file_name): BaseModel
- get_taget_table
- get_taget_forei
- get_taget_requ
+ parse(file_path
+ get_join_confic

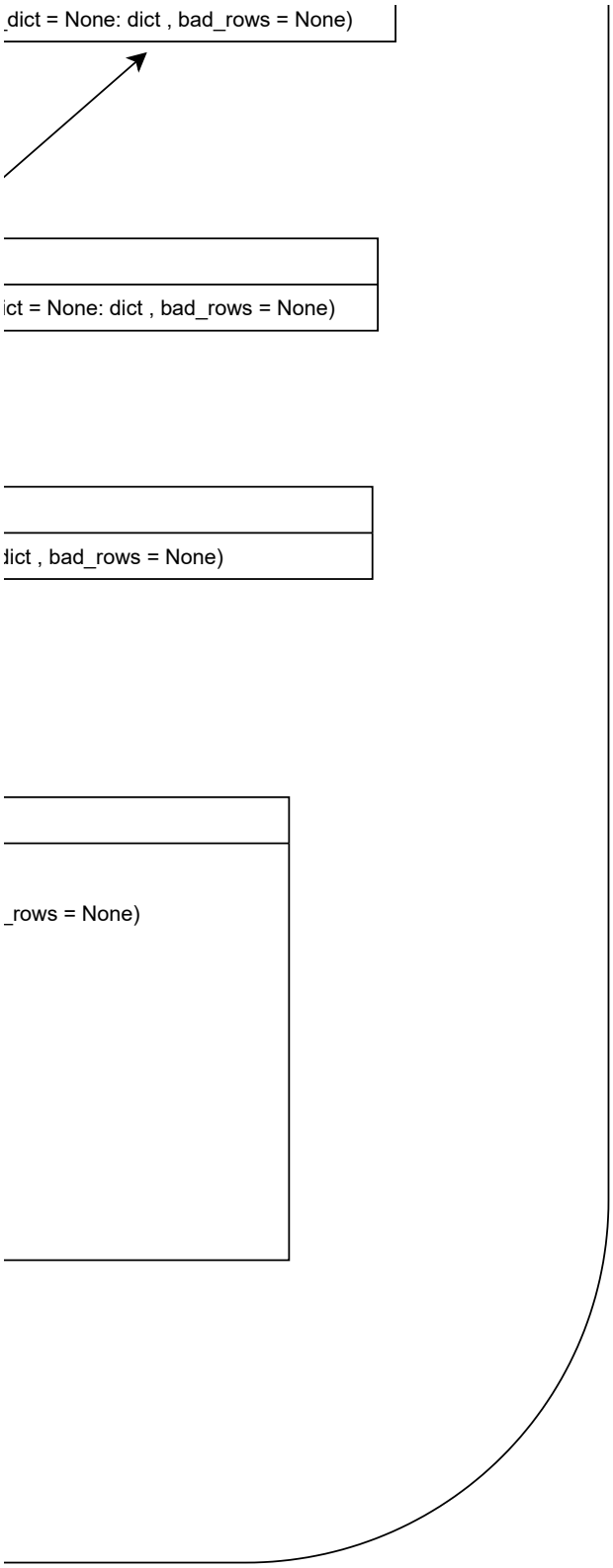

+ validate(df: pd.DataFrame, model: Any, required: list[str]): (bool, list[dict], pd.DataFrame, dict[str, int])
SchemaValidator
+ validate(df: pd.DataFrame, model: Any, required: list[str]): (bool, list[dict], pd.DataFrame, dict[str, int])
ConfFileParser
e_name(conf_section):string
ign_keys(conf_section):string
ired_fields(conf_section):string
n: string): Dict
g(conf_section): List


PIIM
+ do_task(): (bool, List<string> ,data_frame_dict: di
WorkFlow
- batch_mode:string
- validator: ValidatorContext
- registry: DataRegistry
- parser: ConfFileParser
- Audit: Audit
- data_flow_task:DataFowTask
- alerter: EmailTask
+ trigger_alert(message: string): void
+ orchistracte(list<string>): void
DataFlowTasksCreator
+ create_data_flow_task(batch_mode:string):DataFLowTask
+ do_task(): (bool, List<
LookUpFactory
+ create_lookup(type):lookUp
+ do_task(): (bool, List<string> ,dat
+ do_task(): (bool,
BatchModule
BatchReader
work_flow:WorkFlow
audit:Audit
file_tracker:FileTracker


Mask
ict , metrics_dict = None: dict , bad_rows = None)
+ create_source(file_n
LookUp
<string> ,data_frame_dict: dict , metrics_dict: dict , bad_rows)
OrphanLookUp
ta_frame_dict: dict , metrics_dict = None: dict , bad_rows = None)
Orp
- still_orphan (dimName: str , rec_id): bool
- reomve_from_orphans (attribute:string): bool
+ do_task(): (bool, List<string> ,data_frame_dic
DuplicatesLookUp
List<string> ,data_frame_dict: dict , metrics_dict = None: dict , bad_rows = None)
Transformer
+ do_task(): (bool, List<string> ,data_frame_dict: dict , metrics_dict = None: dict , bad_rows = None)
+ remove_unused_columns(): (bool, List<string> , data_frame)
+ calculate_aggregated_columns(): (bool, List<string> , data_frame)

AppManger


ReadFromSourceFactory
ame , DataRegistry , ConfFileParser):ReadFromSource
+ do_task(): (bool, List<string> ,data_frame_dict: dict , metrics_
ReadFromJSON
+ do_task(): (bool, List<string> ,data_frame_dict: dict , metrics_di
ReadFromCSV
+ do_task(): (bool, List<string> ,data_frame_dict: dict , metrics_dict = None: dict , bad_rows = None)
QuarantineWriter
+ do_task(): (bool, List<string> ,data_frame_dict: dict , metrics_dict = None: d
hansHandler
ct: dict , metrics_dict = None: dict , bad_rows = None)
SCDComponent
+ prepare_type_2_new_rows(): void
+ do_task(): (bool, List<string> ,data_frame_dict: dict , metrics_dict = None: dict , bad_
+ handle_scd() : void
+ apply_scd_type_1() : void
+ apply_scd_type_2() : void
+ detect_changed_rows(): void
+ split_new_and_existing_rows(): void
+ generate_surrogate_keys(): void
+ expire_current_rows(): void


_dict = None: dict , bad_rows = None)
ict = None: dict , bad_rows = None)
dict , bad_rows = None)
_rows = None)

	agent_name : VARCHAR(19) NOT NULL
	agent_email : VARCHAR(32) NOT NULL
	agent_phone : INT NOT NULL
FK	team_id : INT NOT NULL
	skill_level : VARCHAR(6) NOT NULL
	hire_date : DATE NOT NULL
	avg_handle_time_min : INT NOT NULL
	resolution_rate : DECIMAL NOT NULL
	csat_score : DECIMAL NOT NULL
	is_active : BOOLEAN NOT NULL
	created_at : TIMESTAMP NOT NULL
	updated_at : TIMESTAMP NOT NULL

	email : VARCHAR(33) NULL
	phone : VARCHAR(11) NULL
	region_id : DECIMAL NULL
FK	segment_id : INT NOT NULL
	signup_date : DATE NOT NULL
	gender : VARCHAR(6) NOT NUI
	created_at : TIMESTAMP NOT N
	updated_at : TIMESTAMP NOT

TicketEvent	
PK	**event_id : UUID NOT NULL**
FK	ticket_id : UUID NOT NULL
FK	agent_id : INT NOT NULL
	event_ts : TIMESTAMP NOT NU
	old_status : VARCHAR(10) NUL
	new_status : VARCHAR(10) NO
	notes : VARCHAR(36) NOT NUI

Driver	
	created_at : TIMESTAMP NOT NULL
	updated_at : TIMESTAMP NOT NULL
PK	**driver_id : INT NOT NULL**
	driver_name : VARCHAR(19) NULL
	driver_phone : VARCHAR(11) NULL
	national_id : INT NOT NULL
FK	region_id : INT NOT NULL
	shift : VARCHAR(7) NOT NULL
	vehicle_type : VARCHAR(9) NOT NULL
	hire_date : DATE NOT NULL
	rating_avg : DECIMAL NOT NULL
	on_time_rate : DECIMAL NOT NULL
	cancel_rate : DECIMAL NOT NULL
	completed_deliveries : INT NOT NULL
	is_active : BOOLEAN NOT NULL

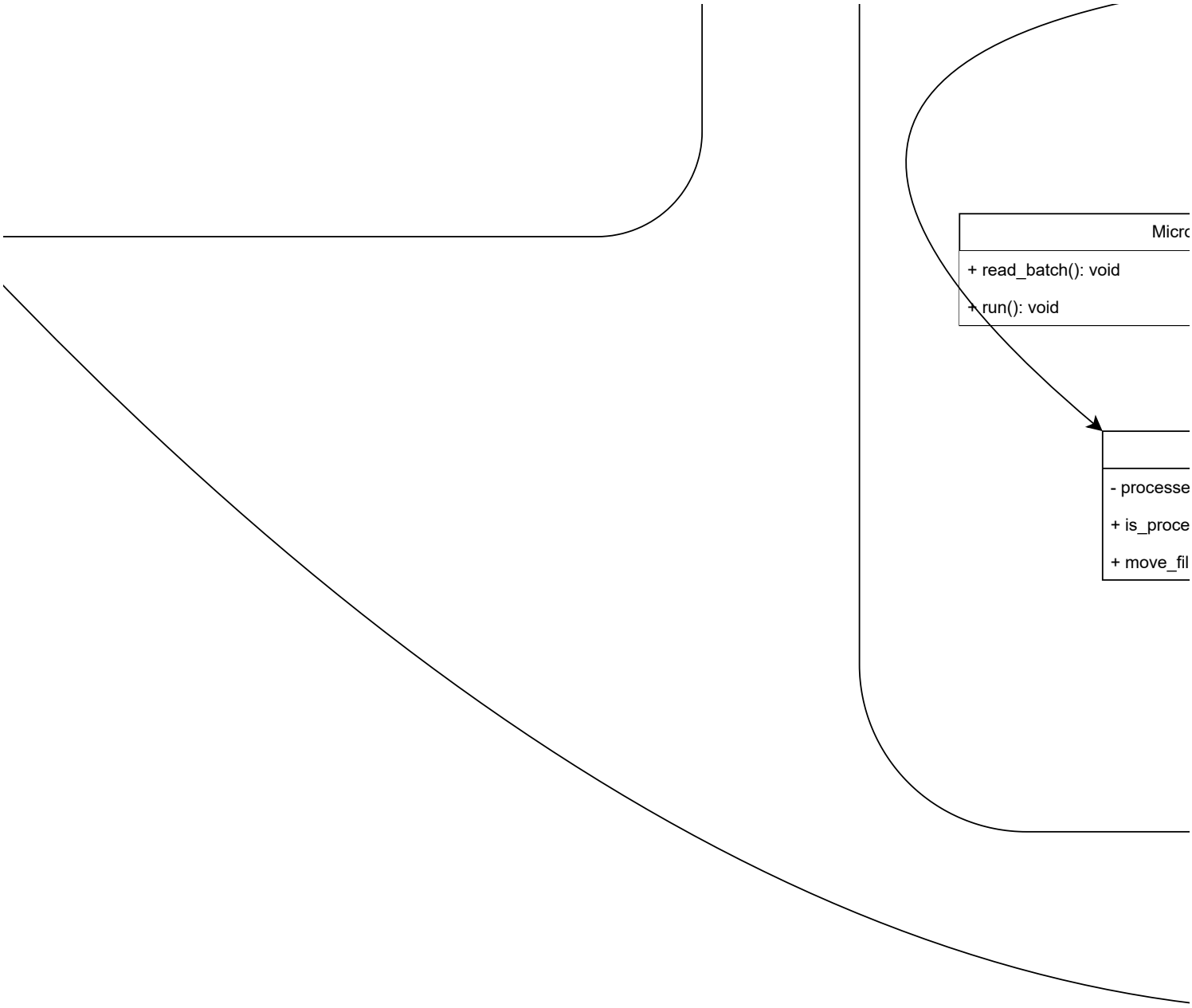
Reason	
PK	**reason_id : INT NOT NULL**
	reason_name : VARCHAR(25) N
FK	reason_category_id : INT NOT N
	severity_level : INT NOT NULL
	typical_refund_pct : DECIMAL N

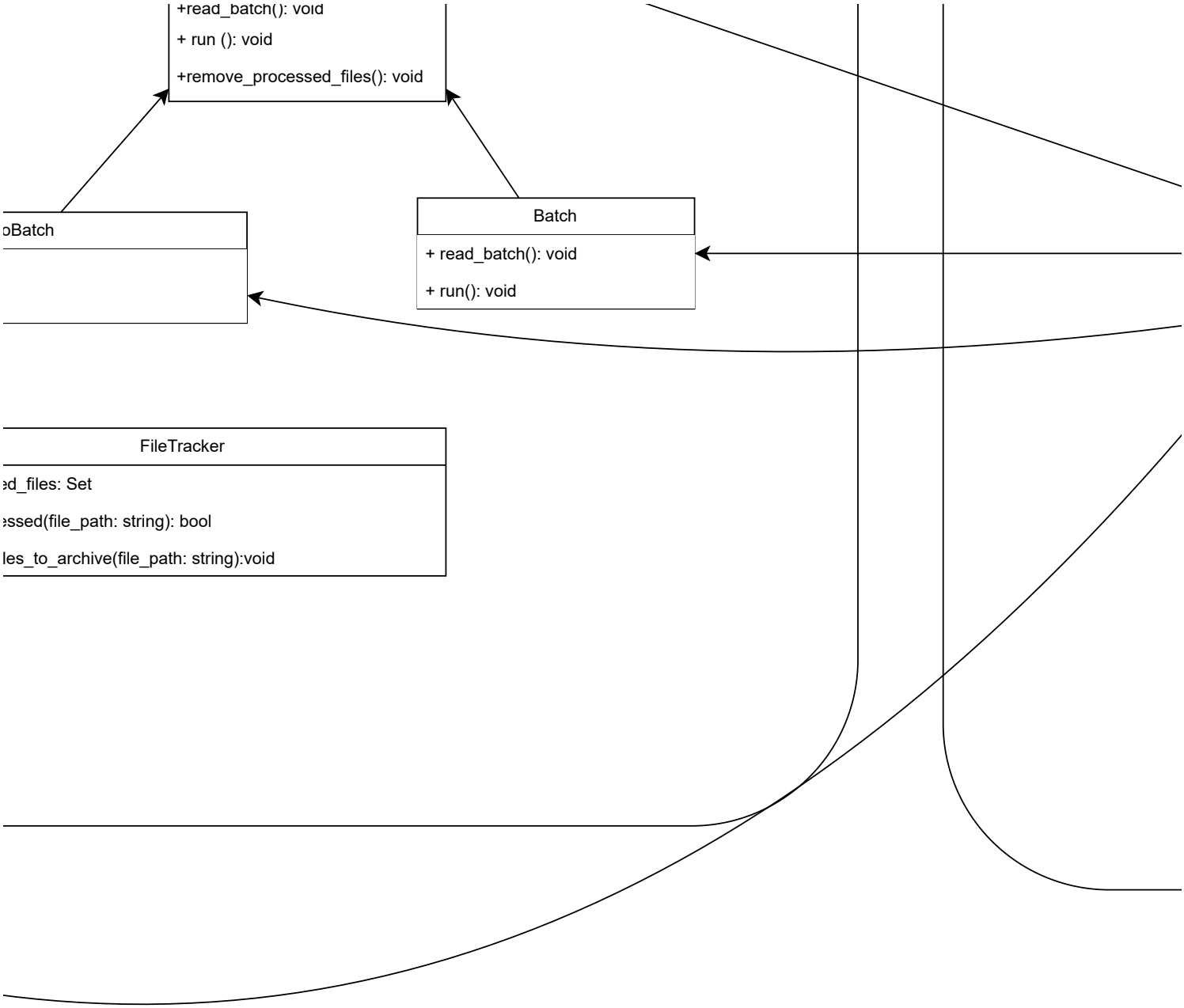
Region	
PK	**region_id : INT NOT NULL**
	region_name : VARCHAR(12) N
FK	city_id : INT NOT NULL
	delivery_base_fee : DECIMAL N

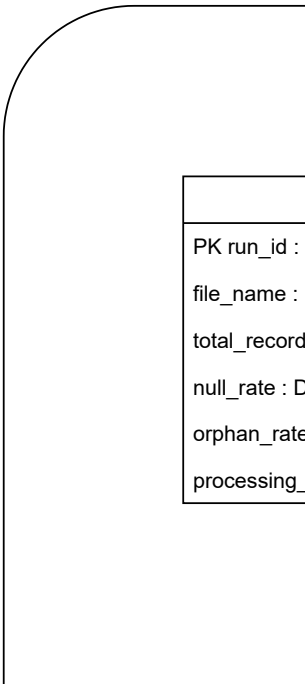
PK	**order_id : UUID NOT NULL**
FK	customer_id : INT NOT NULL
FK	restaurant_id : INT NOT NULL
FK	driver_id : INT NOT NULL
FK	region_id : INT NOT NULL
	order_amount : DECIMAL NOT NULL
	delivery_fee : DECIMAL NOT NULL
	discount_amount : INT NOT NULL
	total_amount : DECIMAL NOT NULL
	order_status : VARCHAR(9) NOT NULL
	payment_method : VARCHAR(6) NOT NULL
	order_created_at : TIMESTAMP NOT NULL
	delivered_at : TIMESTAMP NULL

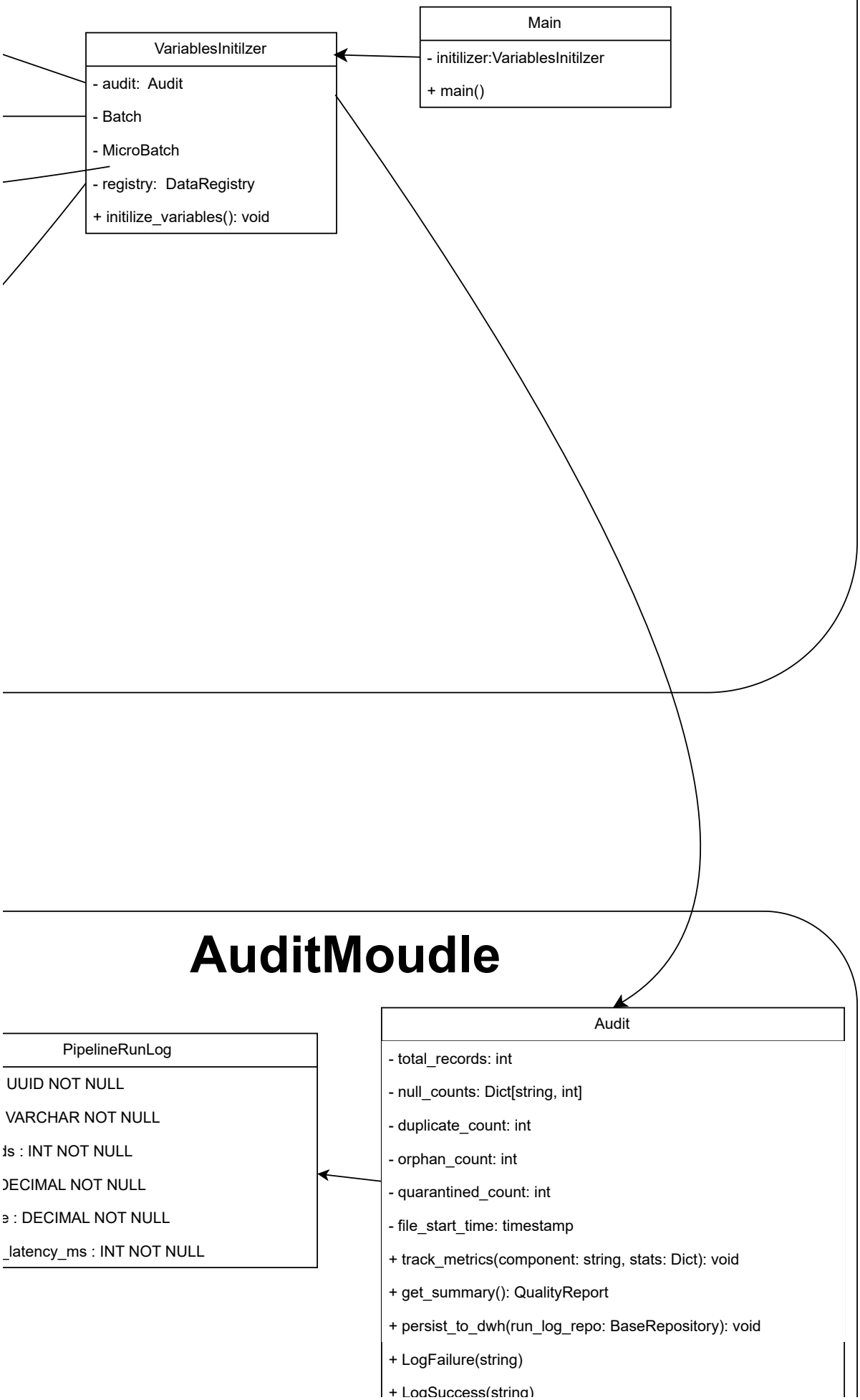
Category	
PK	**category_id : INT NOT NULL**
	category_name : VARCHAR(13) NOT NULL

Restaurant	
PK	**restaurant_id : INT NOT NULL**
	restaurant_name : VARCHAR(19) NULL
FK	region_id : INT NOT NULL
FK	category_id : INT NOT NULL
	price_tier : VARCHAR(4) NOT NULL
	rating_avg : DECIMAL NULL
	prep_time_avg_min : INT NOT NULL
	is_active : BOOLEAN NOT NULL
	created_at : TIMESTAMP NOT NULL
	updated_at : TIMESTAMP NOT NULL


Micro
+ read_batch(): void
+ run(): void
- processe
+ is_proce
+ move_fil


+read_batch(): void
+ run (): void
+remove_processed_files(): void
oBatch
Batch
+ read_batch(): void
+ run(): void
FileTracker
ed_files: Set
essed(file_path: string): bool
les_to_archive(file_path: string):void


PK run_id :
file_name :
total_record
null_rate : D
orphan_rate
processing_


Main
- initilizer:VariablesInitilzer
+ main()
VariablesInitilzer
- audit: Audit
- Batch
- MicroBatch
- registry: DataRegistry
+ initilize_variables(): void
AuditMoudle
Audit
- total_records: int
- null_counts: Dict[string, int]
- duplicate_count: int
- orphan_count: int
- quarantined_count: int
- file_start_time: timestamp
+ track_metrics(component: string, stats: Dict): void
+ get_summary(): QualityReport
+ persist_to_dwh(run_log_repo: BaseRepository): void
+ LogFailure(string)
+ LogSuccess(string)
PipelineRunLog
UUID NOT NULL
VARCHAR NOT NULL
ds : INT NOT NULL
DECIMAL NOT NULL
e : DECIMAL NOT NULL
_latency_ms : INT NOT NULL

+ LogSuccess(string)